

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Naveen

May 2024

Course Code: CIV101

Course: BASIC CIVIL ENGINEERING

QP No. :U038-2

Duration: 3 hours

Max. Marks:100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. A city has a Floor Space Index (FSI) of 2. A developer owns a rectangular plot of land measuring 20 meters by 30 meters. What is the maximum total built-up area allowed for development on this plot?
2. Imagine you're in a construction site and need a quick way to assess the cement before using it. Describe two field tests you could perform and explain what results would indicate good quality cement.
3. Outline the two main types of aggregates you would consider for concrete mix design, classifying them based on their size?
4. Match the following terms with their definitions:

a) Course	i) The adhesive material used to bind bricks together.
b) Mortar	ii) A brick cut to fit at the end of a course.
c) Closer	iii) The specific pattern in which bricks are arranged in a wall.
d) Bond	iv) A horizontal layer of bricks in a wall.

5. Both super elevation and camber involve tilting the road surface, but in different contexts and for varying purposes. Explain the key differences between the two.
6. In the context of choosing spiral stairs for a residential space, outline two key considerations that influence the selection process.
7. While chain surveying has served as a fundamental surveying method for centuries, it comes with its own set of drawbacks. List two limitations of chain surveying.
8. Differentiate between true and magnetic bearings in the context of a surveying compass.
9. Name two factors that affect the choice of irrigation method for a specific agricultural field.
10. Identify the structure in dams responsible for managing the controlled release of water and explain its purpose.

PART – B

Answer all the questions

4 x 15 = 60 Marks

11. You are an architect designing a new office building for a rapidly growing tech company with 500 employees. The client desires an innovative and sustainable design. You need to outline your approach to building planning, considering key principles for a successful design and construction.

(OR)

12. a) During a field trip to an ancient urban area, architecture students were assigned to compare buildings constructed with traditional stone masonry and brick masonry in terms of durability, aesthetics, and sustainability. As a student, outline the main differences in these aspects in a simple and brief manner. (10)

- b) State the chemical compounds responsible for the setting and hardening of cement. Briefly explain their role in the cement hydration process. (5)

13. As a civil engineer designing a foundation for a new two-story house on expansive soil, suggest the most suitable five distinct foundation types, accompanied by informative sketches illustrating their applicability to challenging expansive soil conditions.

(OR)

14. As a designer tasked with selecting doors based on functionality and space utilization, provide recommendations and justifications with accompanying sketches for the following scenarios:

- Welcoming Storefront for a High-End Retail Store.
- Renovation of a Historic Building.
- Practical and Affordable Doors for a Hospital.
- Secure Closure for a Warehouse.
- Restaurant facilitates both air exchange and privacy.

15. As a land surveyor heading a team for a complex urban development project, recommend surveying instruments tailored for diverse scenarios, including residential, commercial, and recreational areas, with varying terrain and existing infrastructure. Provide justifications for each instrument choice.

(OR)

16. The following readings were taken with a levelling instrument with a bench mark RL = 81m, the instrument was shifted after 3, 7, 9th reading. 0.320, 0.990, 0.770, 1.030, 1.105, 1.110, 0.550, 1.400, 1.000, 0.330, 1.200. Find the RL of all the points using the rise and fall method.

17. In a village experiencing a decline in groundwater levels due to prolonged monsoon failure, the local panchayat aims to address water scarcity for drinking and irrigation. Identify the most suitable rainwater harvesting methods for individual households and illustrate their components with clear sketches.

(OR)

S.P

18. In response to an imminent flood threat from a sudden storm, an emergency flood control team is tasked with deploying non-rigid dams due to limited resources and time constraints. Explain various types of non-rigid dams, accompanied by neat sketches, considering the urgency of the scenario.

PART – C

Answer the following

1 x 20 = 20 Marks

19. a) Provide recommendations for the use of specific commercial forms of steel and timber in the construction of a multi-purpose commercial complex, emphasizing structural requirements and environmental sustainability. (10)
- b) Additionally, express your preference between framed structures and load-bearing structures, justifying your choice based on the unique characteristics and needs of the commercial complex. (10)
